

STAT 312: HW Assignment 1 - Unit 01 & 02

Instructor: Dr. Md Mutasim Billah

Submission Guidelines

Please follow these guidelines carefully when completing and submitting your homework. Clear and well-organized work helps ensure fair and accurate grading.

- Submit all homework assignments through **Canvas**.
- If you prefer to work with **pen and pencil**:
 - Download and/or print the homework assignment.
 - Scan your completed work and combine all pages into a **single PDF file**.
 - Upload the PDF to the corresponding homework assignment on Canvas.
- If you prefer to use a **writing device** or type your solutions, you may complete the homework electronically and submit your final work directly as a **PDF**.
- You may also need to submit an **R file** if the homework includes a programming task.
- Your work must be **clear, organized, and legible**. Please follow these presentation guidelines:
 - Clearly circle or otherwise mark your final answers so they can be easily located on the page.
 - Show all necessary work for **nontrivial calculations**. Answers without sufficient supporting work may receive little or no credit.
- Name your submitted PDF using the following format:
`firstname_lastname_HW#.pdf`
For example: `john_smith_HW#.pdf`

Problem 1.1 - Population and Sample

A historian is studying bronze church bells that were manufactured at Hammerhead Foundry in Pennsylvania during the 1840s. These bells have a reputation for being very loud, and he is interested in exactly how loud they are. He has accumulated a list of all such bells that are still in use, and will randomly select 15 of them to visit in person.

- (a) What is the population?
- (b) What is the sample?
- (c) The historian visits the 15 bells he randomly selected and measures their loudness in decibels (dB) when rung. The data are:

90.7	93.3	93.5	98.2	96.3
94.4	90.7	93.3	96.8	90.3
98.6	98.5	96.2	91.5	94.1

Draw a histogram using the following class intervals: $[90, 92)$, $[92, 94)$, $[94, 96)$, ... etc. First, construct a frequency table containing the class intervals, frequencies, and relative frequencies. Then draw the corresponding histogram. Describe the histogram as right-skewed, left-skewed, or symmetric.

Problem 1.2 - Stem-and-Leaf Plots

Biologists are studying a woman who has given birth to twenty children (ten boys and ten girls). When each child reached 10 years of age, their height (in inches) was recorded:

Boys:

51.3, 52.8, 53.4, 53.5, 54.1, 54.6, 54.6, 55.2, 55.7, 55.8

Girls:

50.5, 51.0, 51.4, 52.7, 52.8, 53.1, 53.6, 54.4, 54.4, 55.3

- (a) Use these data to draw side-by-side stem-and-leaf plots. Have your stems be 50, 51, 52, 53, 54, 55. Use $50 \mid 5 = 50.5$ as the legend.
- (b) Describe the plots as right-skewed, left-skewed, or symmetric.

Problem 1.3 - Numerical Summaries & Boxplot

A professor teaches a class with 51 students. The maximum possible point total for the semester is 1,000. At the end of the semester, the professor orders the data from least to greatest. For example, the student with the lowest point total is 64 and the student with the highest point total is 998.

1. 64	18. 659	35. 824
2. 365	19. 679	36. 831
3. 544	20. 686	37. 836
4. 564	21. 699	38. 844
5. 581	22. 701	39. 857
6. 582	23. 703	40. 861
7. 584	24. 704	41. 865
8. 605	25. 708	42. 868
9. 608	26. 718	43. 897
10. 612	27. 725	44. 901
11. 616	28. 726	45. 932
12. 622	29. 740	46. 941
13. 623	30. 755	47. 953
14. 628	31. 789	48. 954
15. 633	32. 796	49. 957
16. 647	33. 796	50. 985
17. 652	34. 811	51. 998

- (a) For this data set, find the values of the lower quartile (Q_1), median (Q_2), upper quartile (Q_3), Interquartile range (IQR), Step, UIF, LIF, UOF, and LOF. *Note:* It is okay if any of the fence values are less than 0 or greater than 1000.

- (b) Using the information you obtained in Problem 1.3(a), draw a boxplot.

Problem 1.4 - Numerical Summaries

- (a) Each of the following data sets has a mean of 10. Without using a calculator, arrange the data sets in order of increasing variance (or standard deviation).

A: {4, 6, 8, 10, 12, 14, 16}

B: {-2, 3, 4, 6, 10, 19, 30}

C: {9, 10, 10, 10, 10, 10, 11}

D: {9, 9, 9, 10, 11, 11, 11}

- (b) Find the sample variance and sample standard deviation of the following data set by hand:

$$\{-2, 3, 4, 6, 10, 19, 30\}.$$

Use both the “regular” formula and the “convenient” formula to calculate the sample variance. Show your work and verify that both formulas give the same answer.

(c) The data set

$$\{0, 2, 3, 3, 4, 6\}$$

has a sample mean of 3, a sample variance of 4, and a sample standard deviation of 2. Using this information - and performing very few additional calculations - find the sample mean, sample variance, and sample standard deviation of each of the following data sets:

A: $\{3000, 3002, 3003, 3003, 3004, 3006\}$

B: $\{0, 10, 15, 15, 20, 30\}$

C: $\{0, 0.2, 0.3, 0.3, 0.4, 0.6\}$

D: $\{-10, -12, -13, -13, -14, -16\}$

GOOD LUCK!